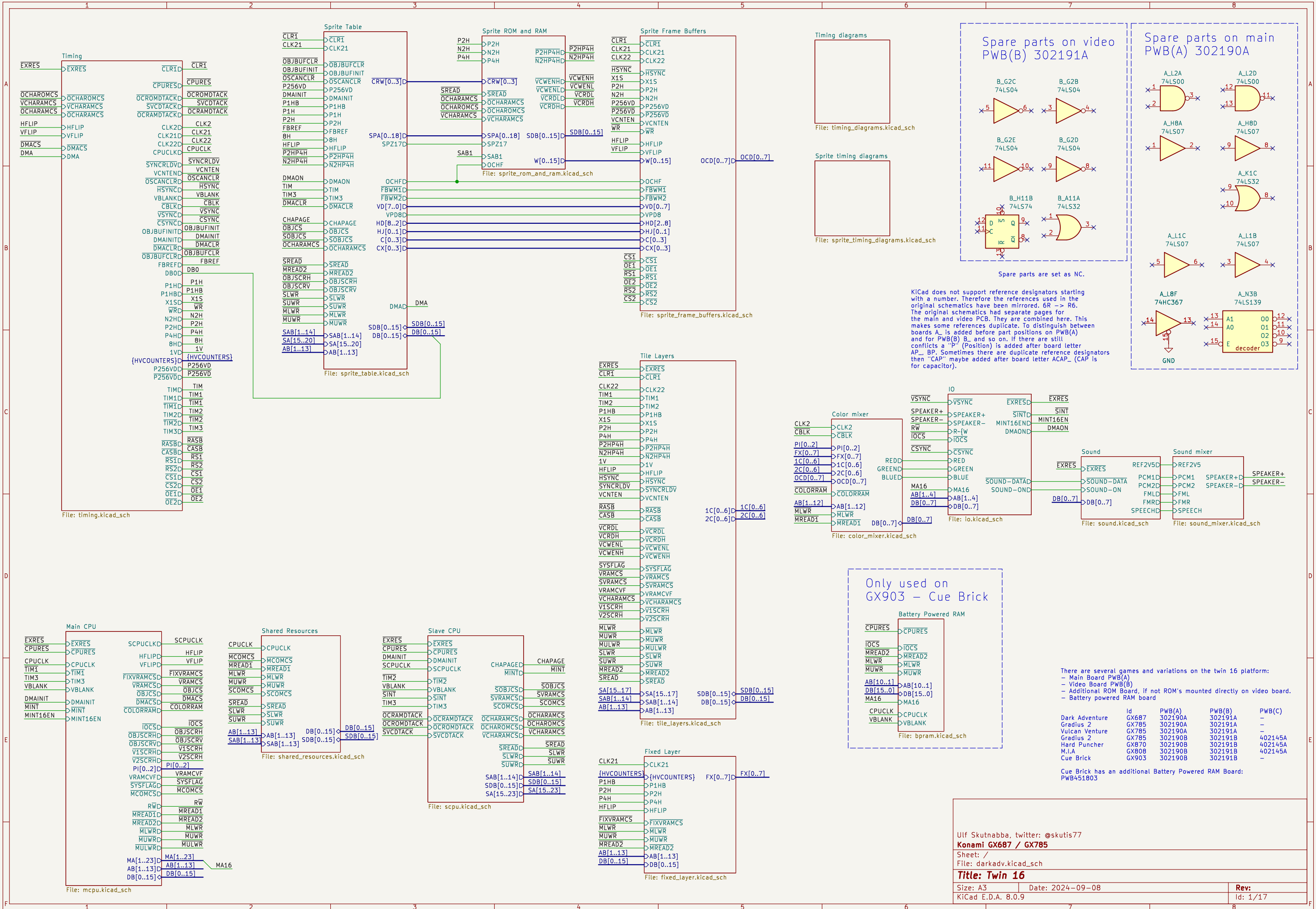
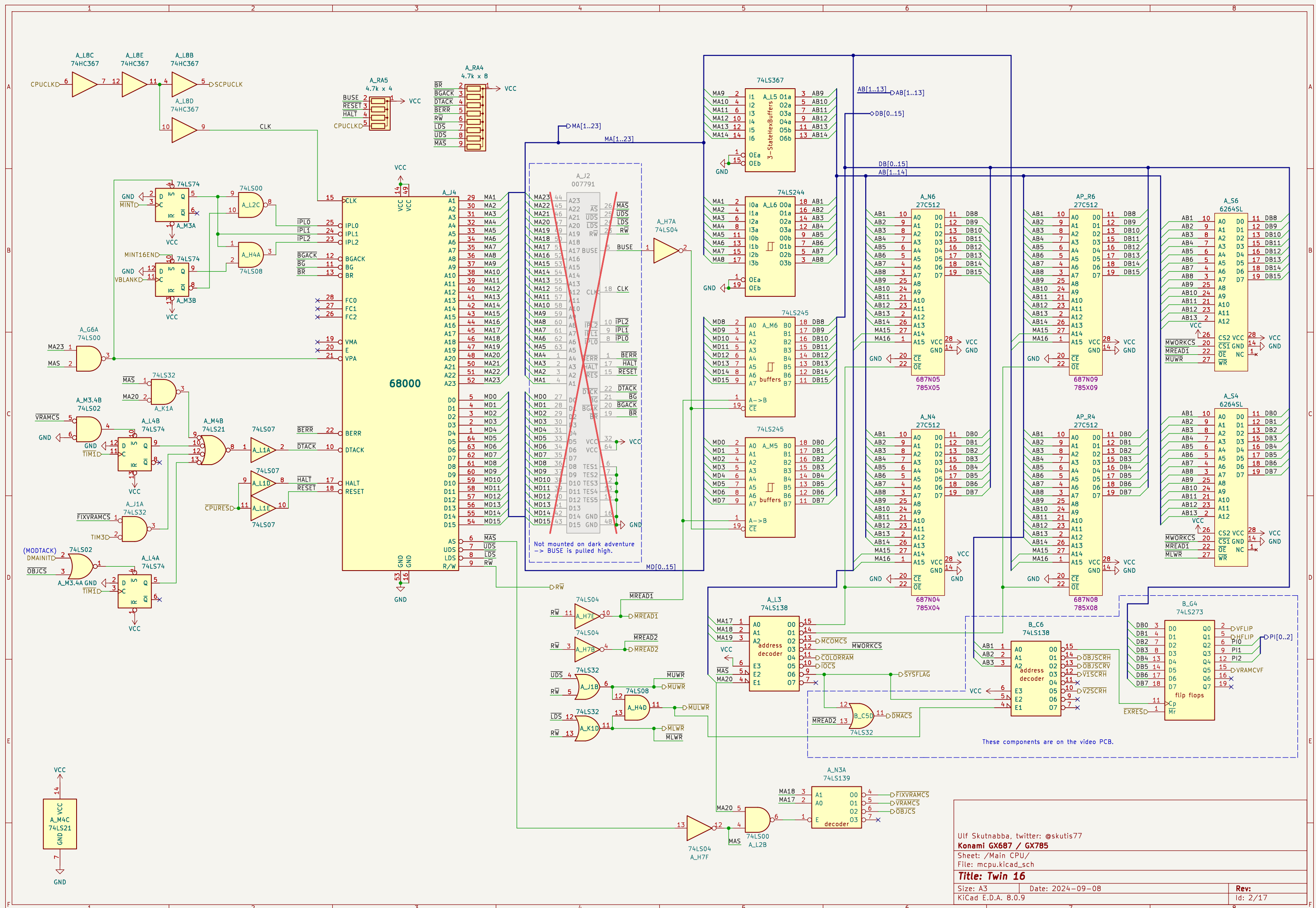
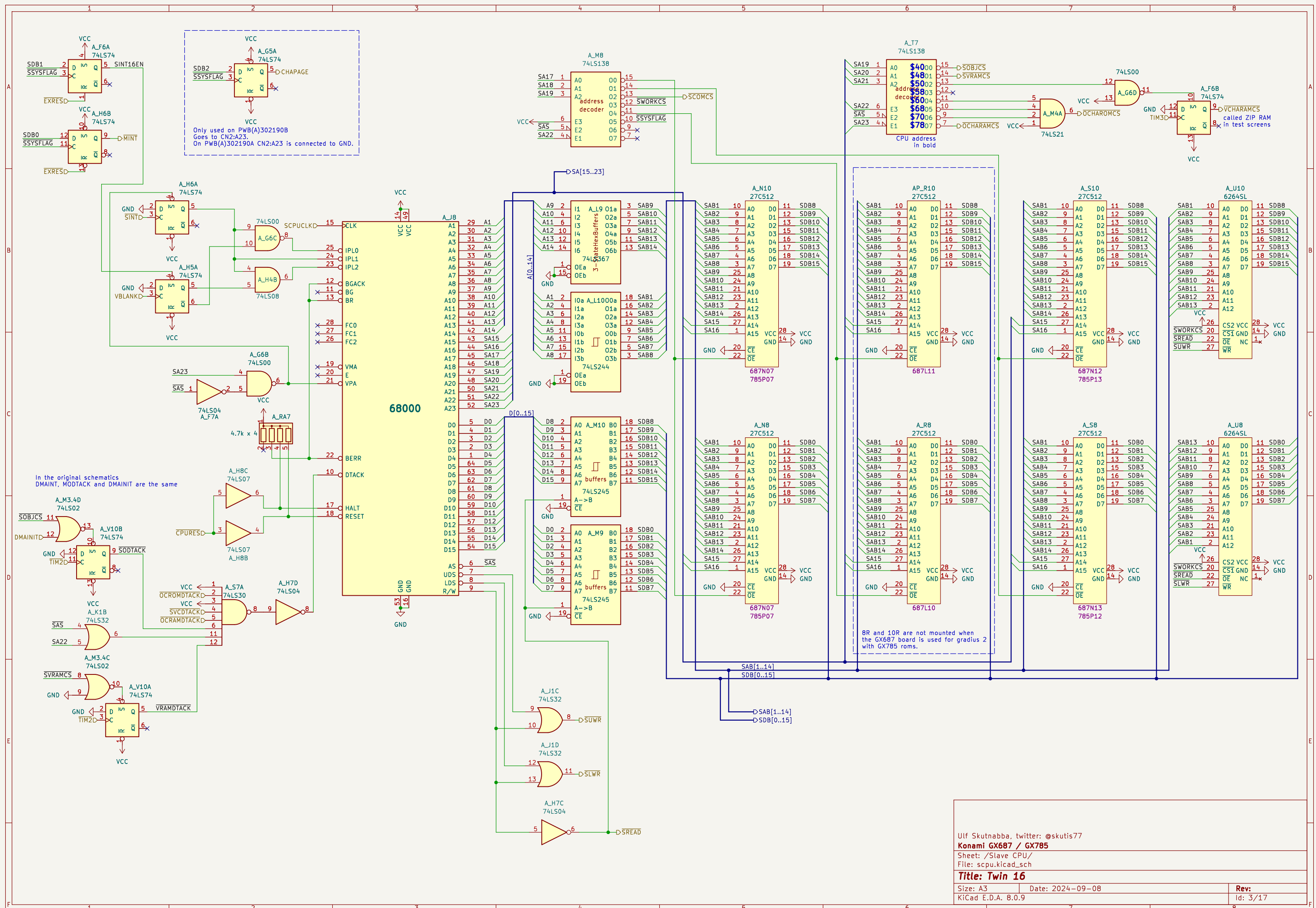




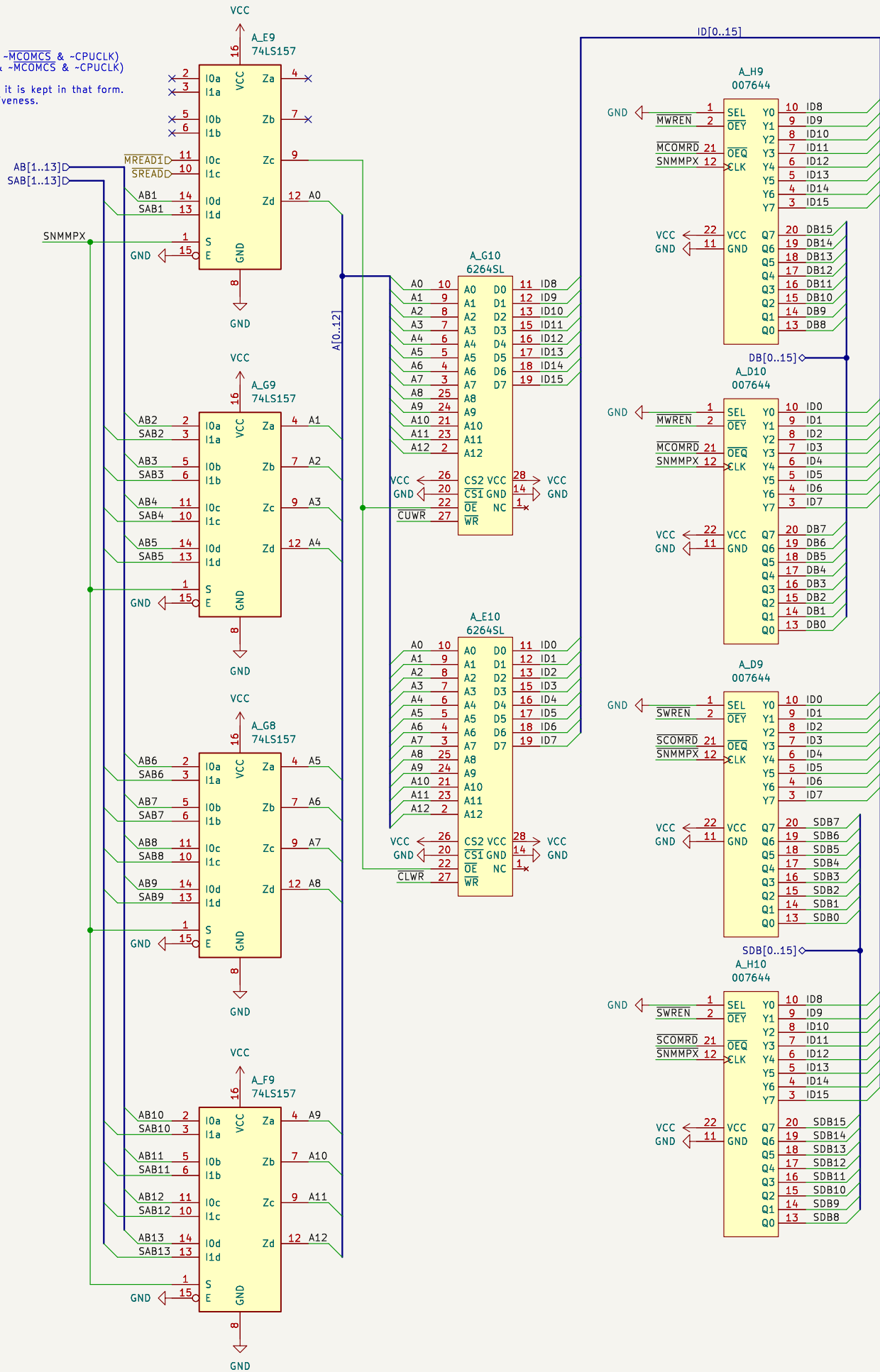
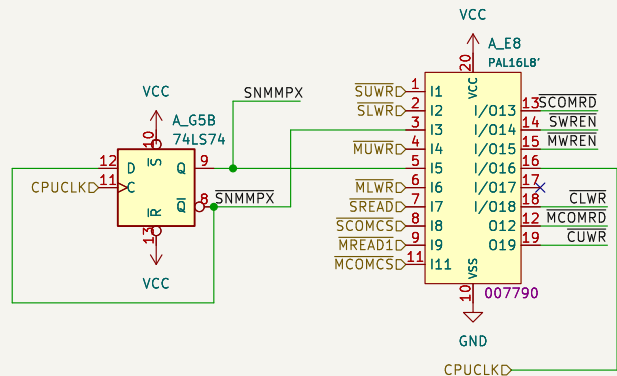




Konami 007790

$\overline{MCOMRD} = \neg(\neg \overline{MREAD1} \& \neg \overline{MCOMCS}) = \overline{MREAD1} \mid \overline{MCOMCS}$
 $\overline{SCOMRD} = \neg(\neg \overline{SREAD} \& \neg \overline{SCOMCS})$
 $\overline{SWREN} = \neg(\neg \overline{SNMMPX} \& \overline{SREAD} \& \neg \overline{SCOMCS})$
 $\overline{MWREN} = \neg(\neg \overline{SNMMPX} \& \overline{MREAD1} \& \neg \overline{MCOMCS})$
 $\overline{CLWR} = \neg(\neg \overline{SLWR} \& \neg \overline{SNMMPX} \& \overline{SNMMPX} \& \overline{SREAD} \& \neg \overline{SCOMCS} \& \neg \overline{CPUCLK} \mid \neg \overline{SNMMPX} \& \neg \overline{MLWR} \& \overline{MREAD1} \& \neg \overline{MCOMCS} \& \neg \overline{CPUCLK})$
 $\overline{CUWR} = \neg(\neg \overline{SUWR} \& \neg \overline{SNMMPX} \& \overline{SNMMPX} \& \overline{SREAD} \& \neg \overline{SCOMCS} \& \neg \overline{CPUCLK} \mid \neg \overline{MUWR} \& \neg \overline{SNMMPX} \& \overline{MREAD1} \& \neg \overline{MCOMCS} \& \neg \overline{CPUCLK})$

All AND logic can be converted to OR form. But since AND logic and inverters are used in the PAL16L8 device it is kept in that form. The equations follow verilog syntax. The active low signals are not inverted, the bar is there only to show activeness.



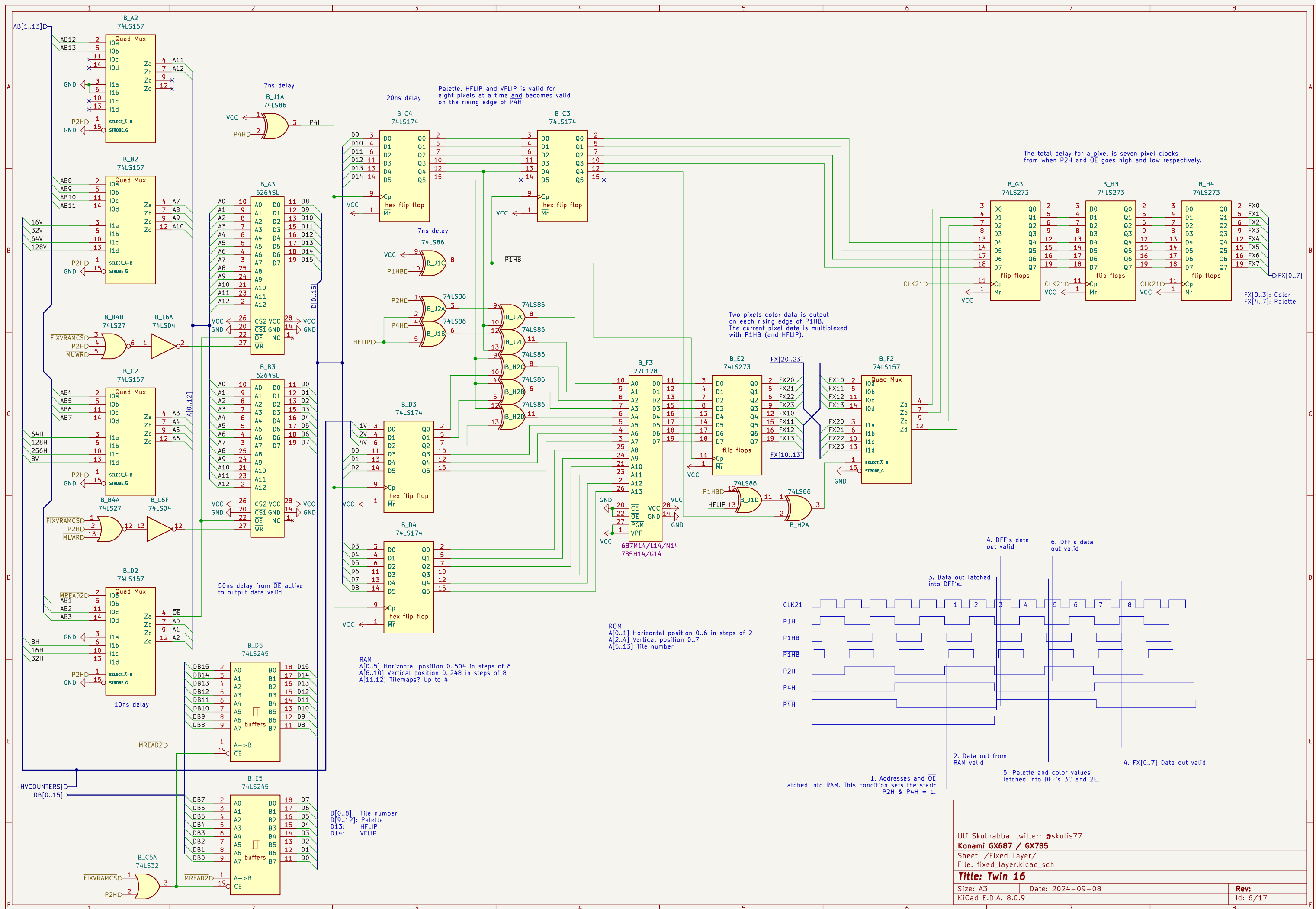
This is from jammarcade.net

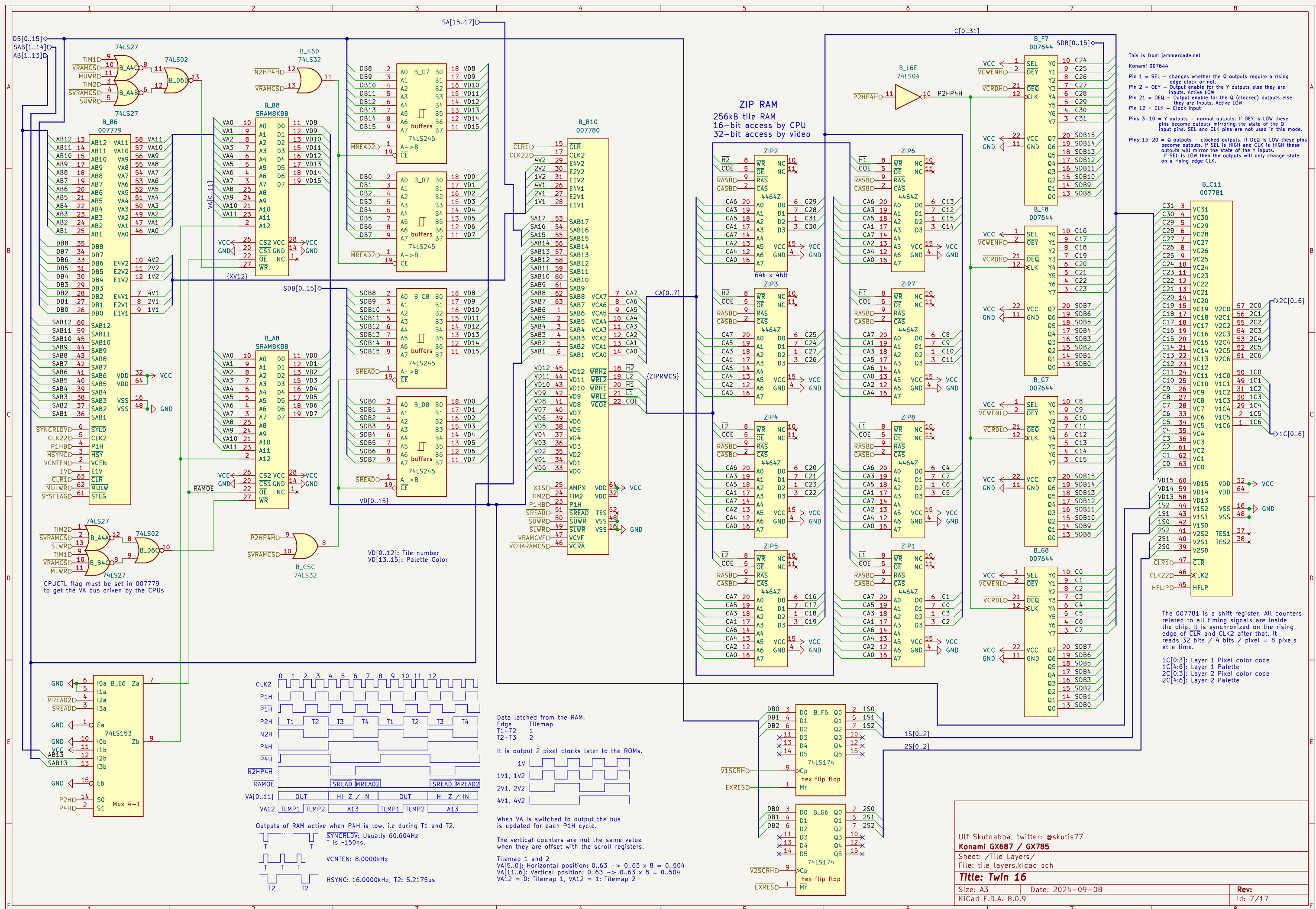
Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

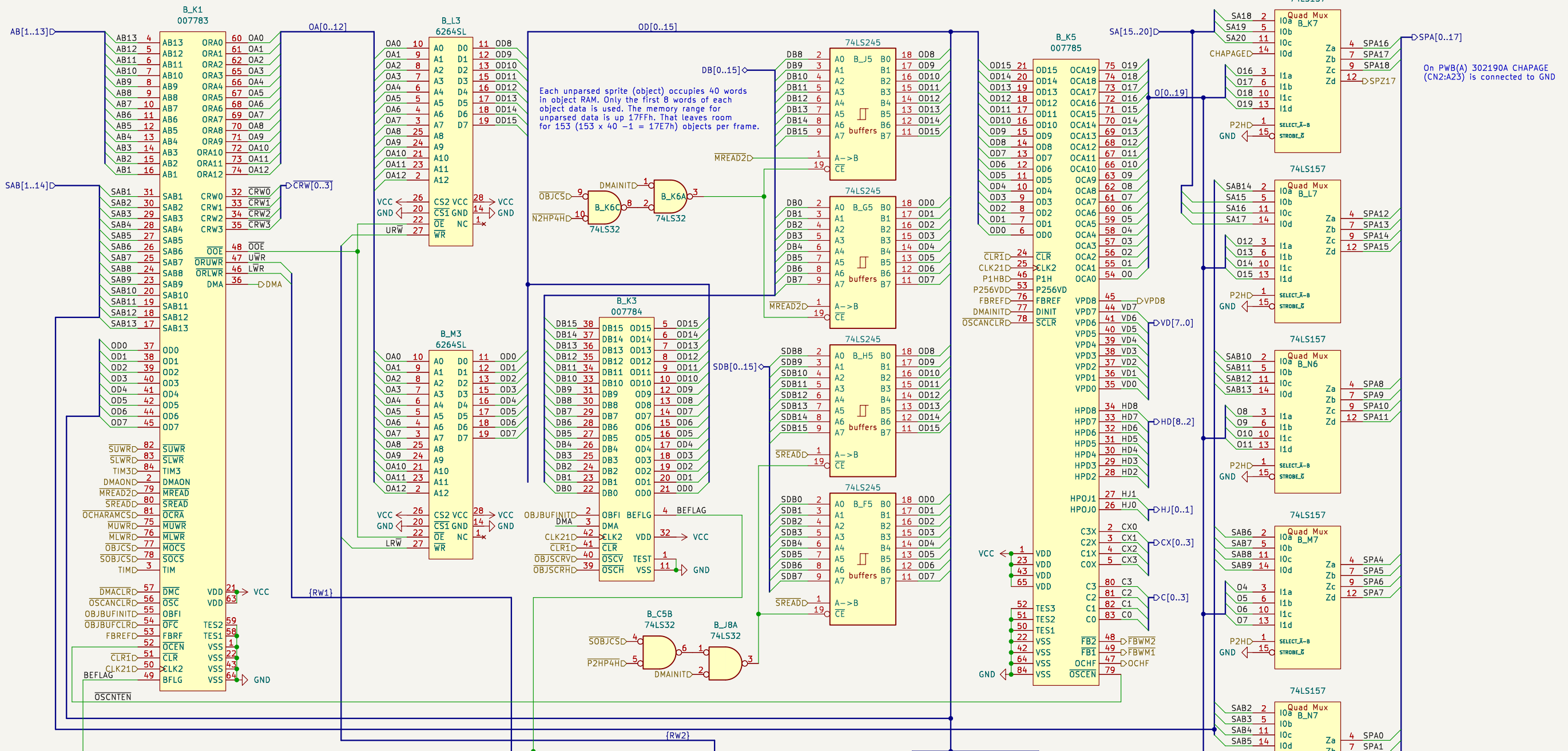
Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs. If SEL is LOW then the outputs will only change state on a rising edge CLK.









007784 Sprite parser

- subtracts the scroll value for the sprites from current coordinates for the sprites.
- When DMA goes high. The 007784 will read 8 words, if bit OD15 of the first word is high then BEFLAG goes high and stays high until the sprite RAM has processed the first four words and possibly updated the four following words at the high end of the RAM. It will compress the eight words into four words and also subtract the x- and y-scroll values from the x and y coordinates. It will write the four words at the high end of the sprite RAM.
- Addresses at the low end goes from 0x0000..0x0007 and the following cycle at 0x0028..0x002F..

Scroll Register

OBJSCRV Sprite horizontal scroll

OBJSCRH Sprite vertical scroll

Data is latched in the scroll registers when the signals OBJSCRV and OBJSCRH go low.

007793

CLK21' = CLK21

LRW = -(DMAONS & OBJBUFINIT & CLK21' | BEFLAG & DMA & P1H & DMAONS & CLK21' & (-BH & HFLIP | 8H & -HFLIP) -LWR & -DMAINIT)

URW = -(DMAONS & OBJBUFINIT & CLK21' | BEFLAG & DMA & P1H & DMAONS & CLK21' & (BH & -HFLIP | -BH & HFLIP) -UWR & -DMAINIT)

The sprite memory is controlled in three separate cases:

1. OBJBUFINIT
2. BEFLAG & DMA
3. -DMAINIT

Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /Sprite Table/

File: sprite_table.kicad_sch

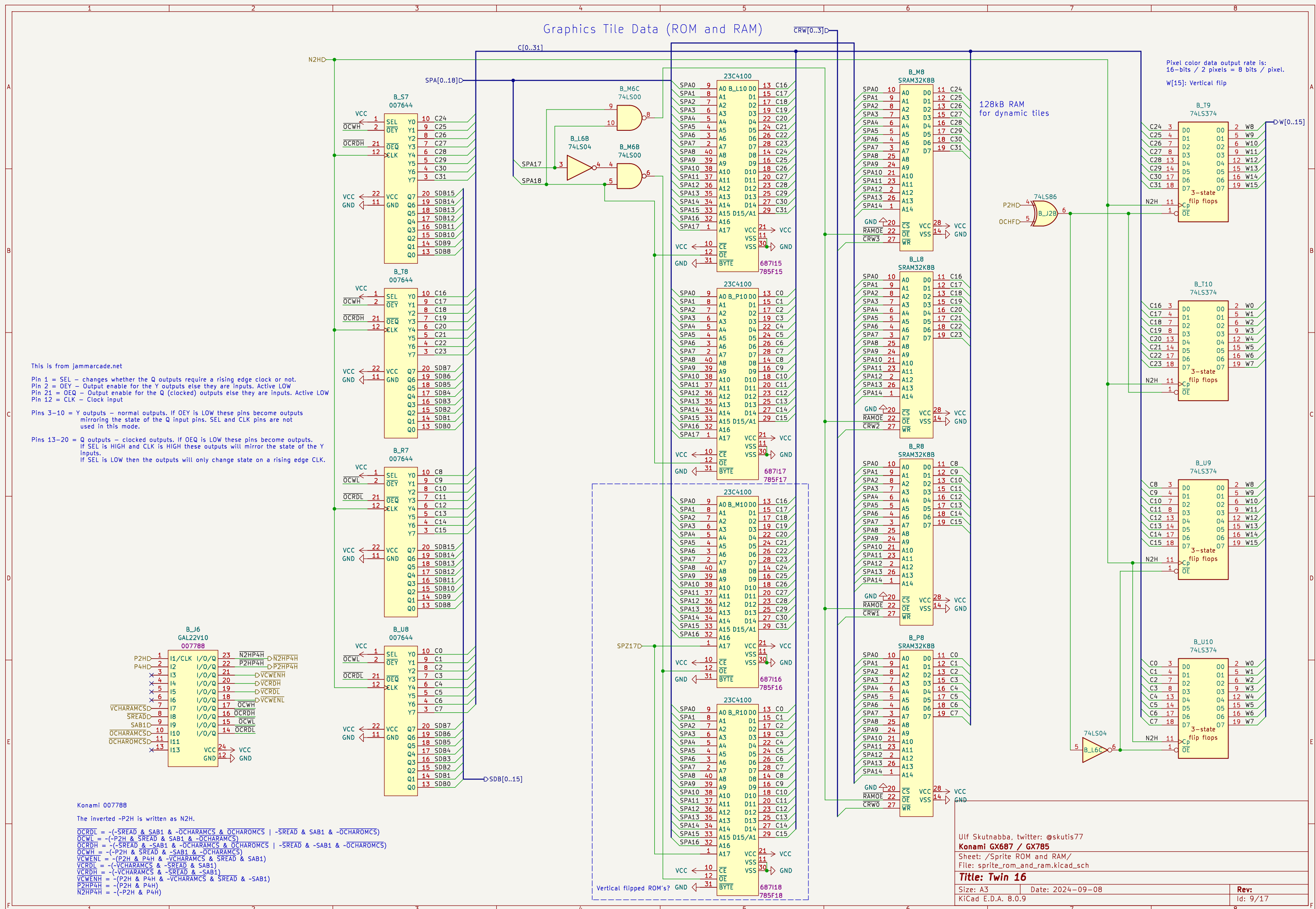
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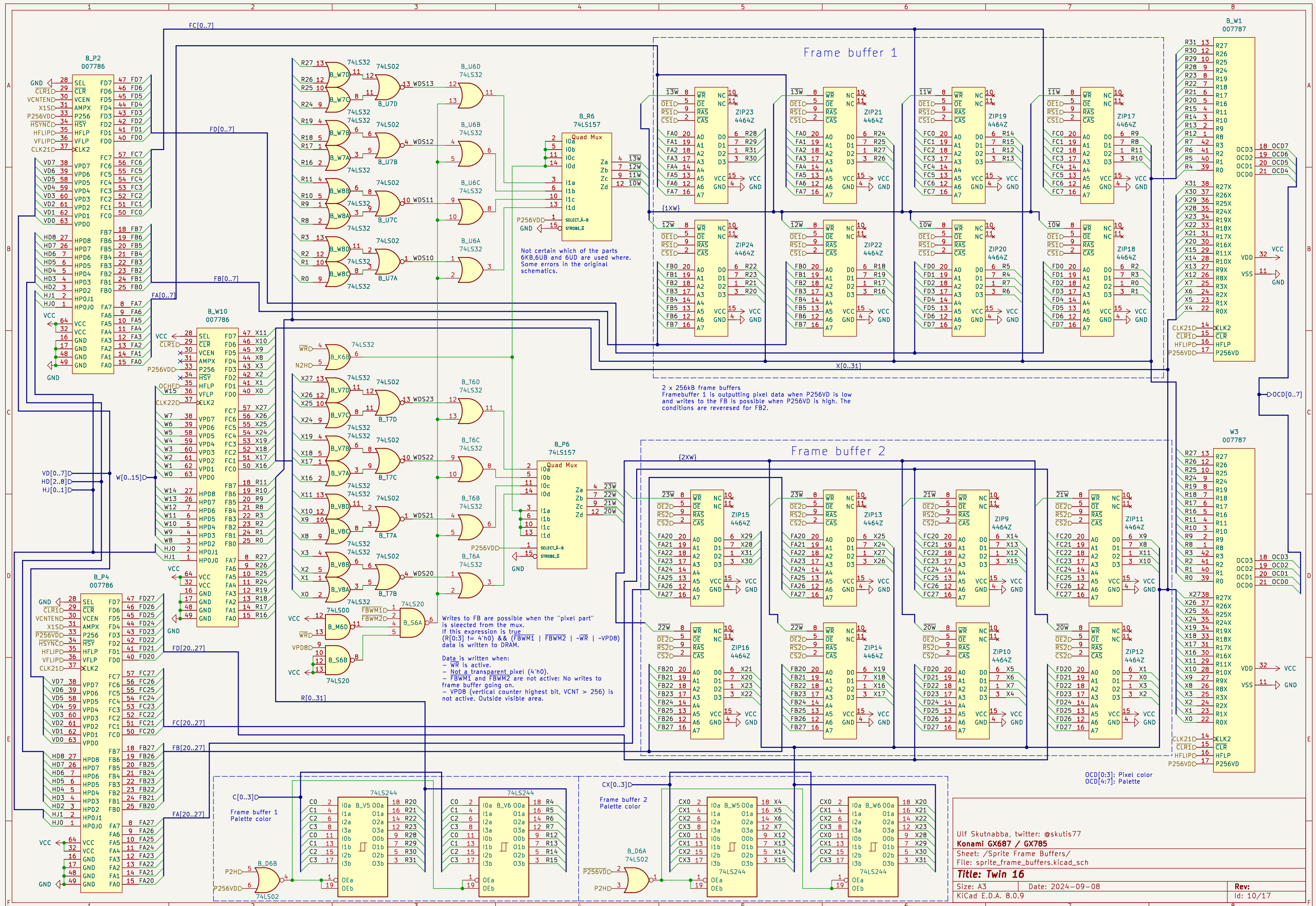
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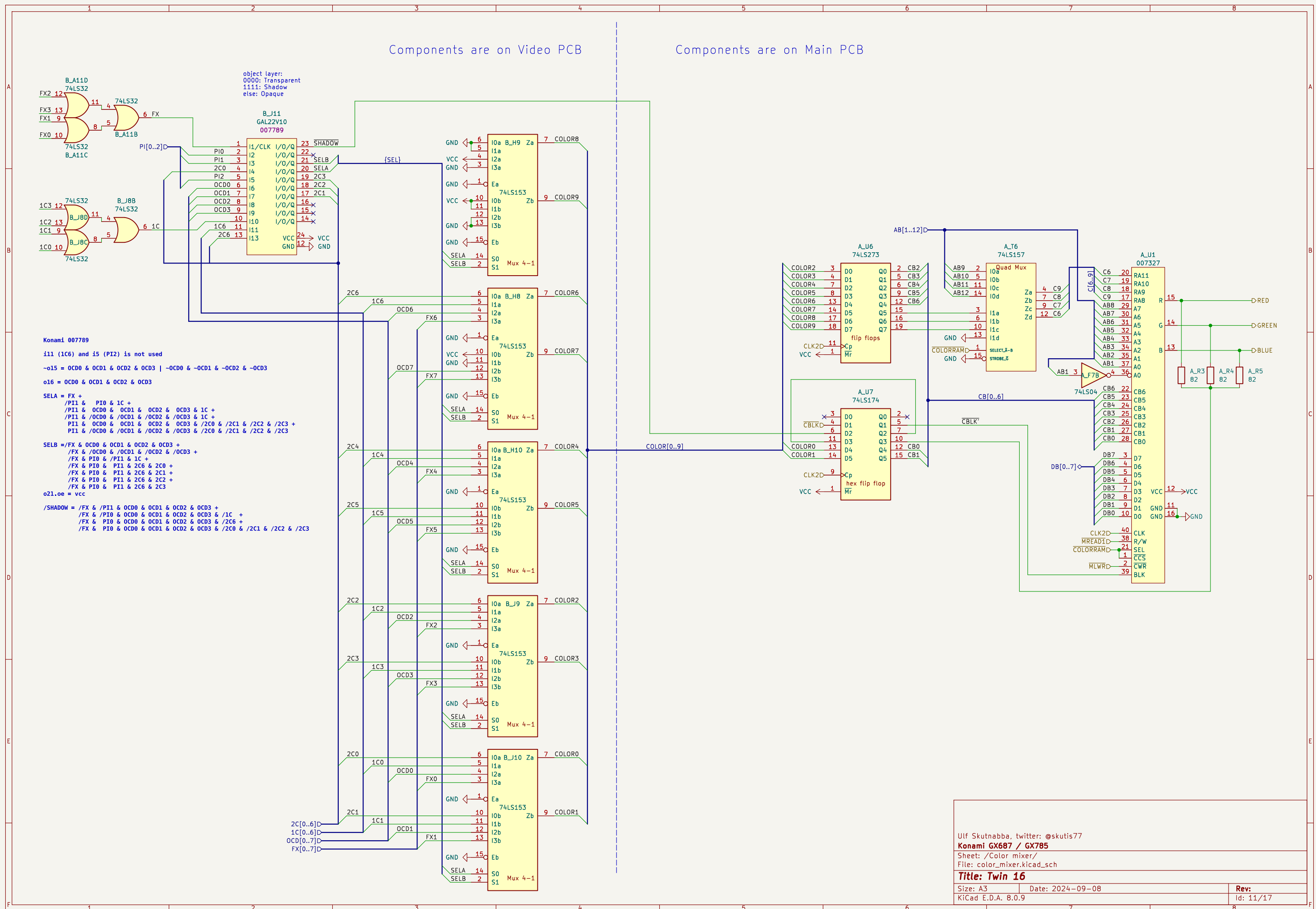
KiCad E.D.A. 8.0.9

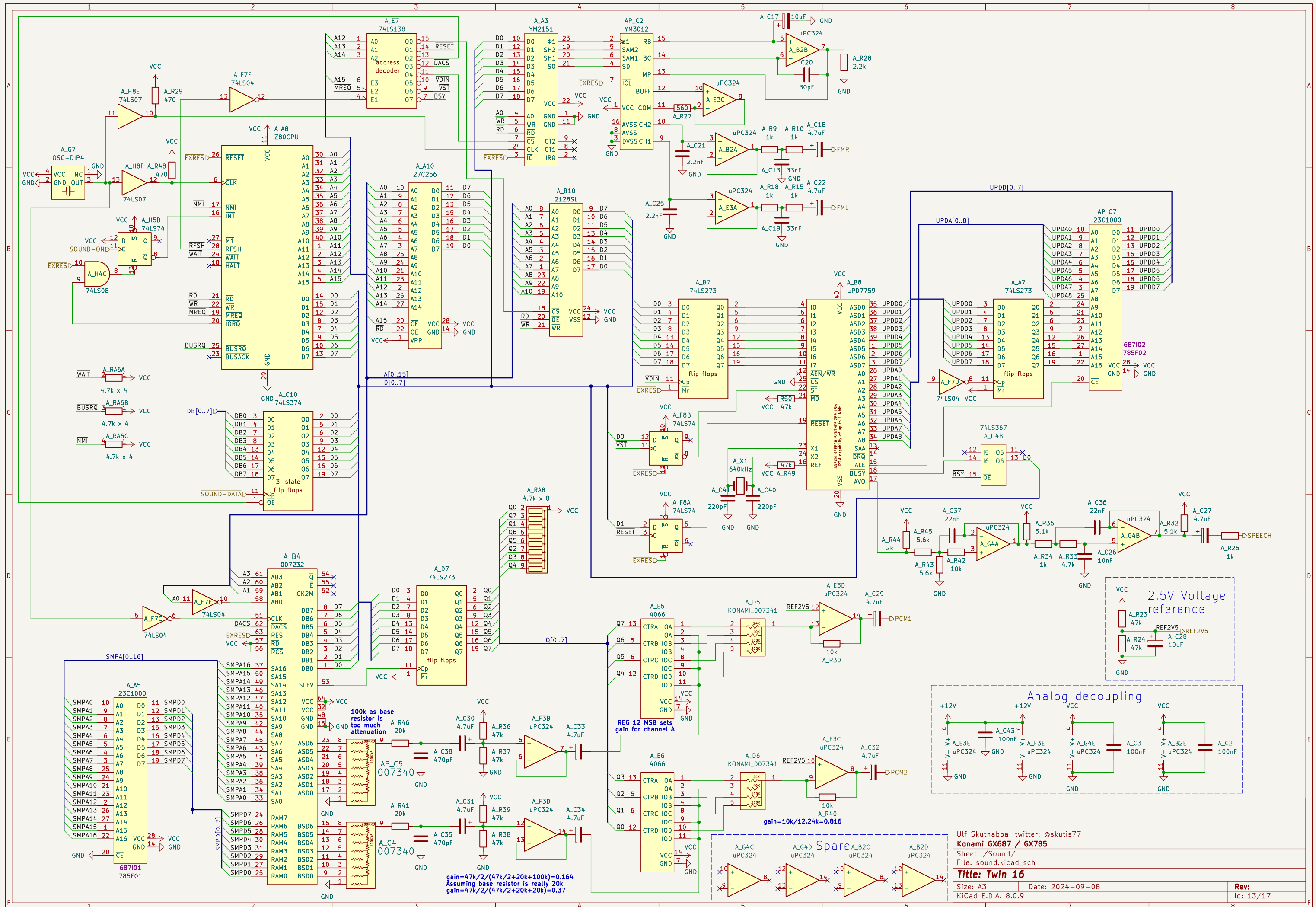
Rev:

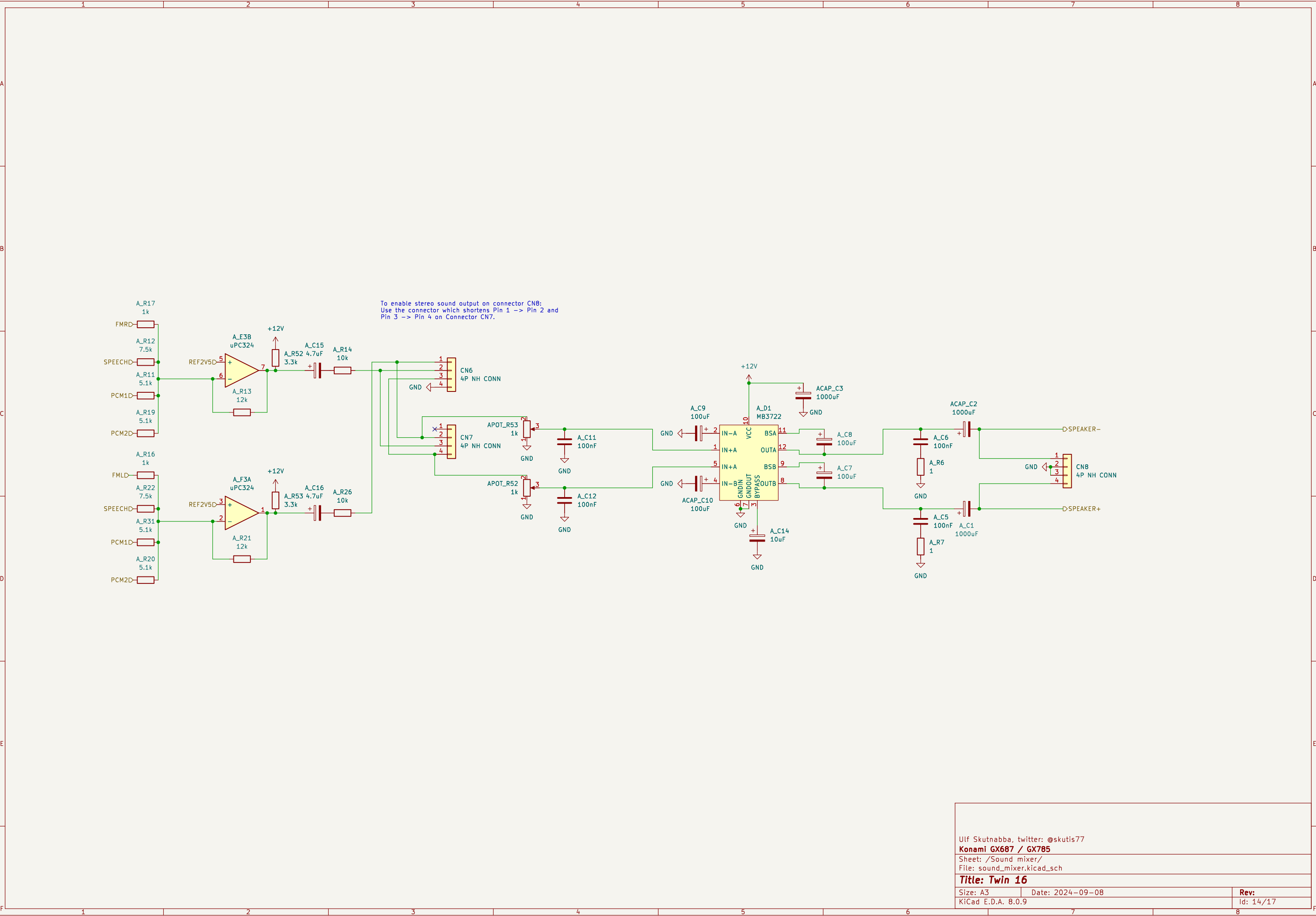
Id: 8/17











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Konami GX687 / GX785

Sheet: /Sound mixer/

File: sound_mixer.kicad_sch

Title: Twin 16

Size: A3

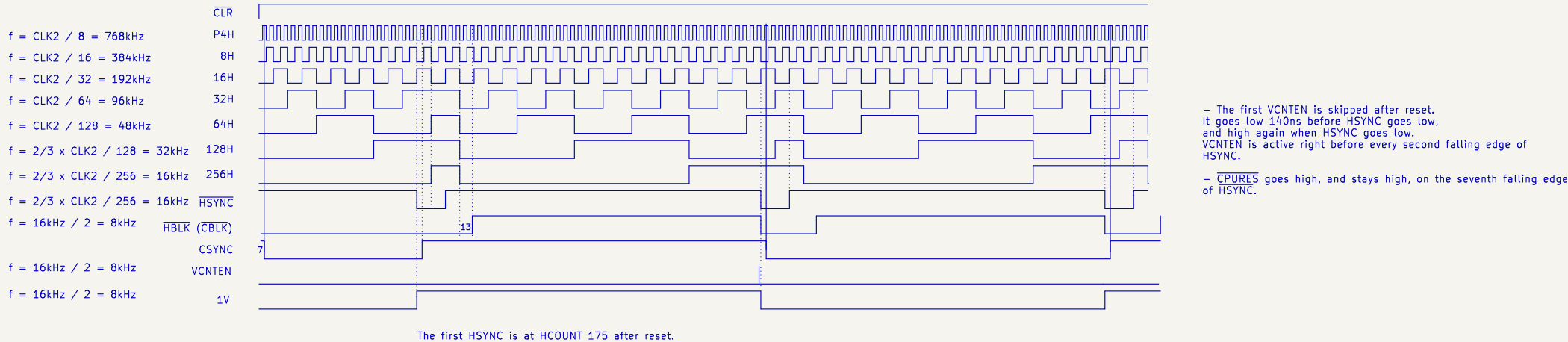
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Rev:

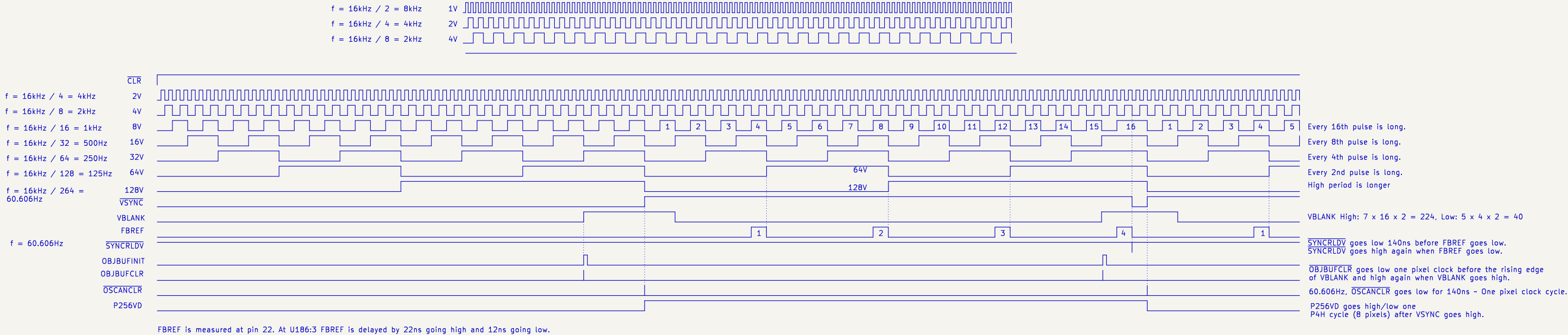
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Id: 14/17

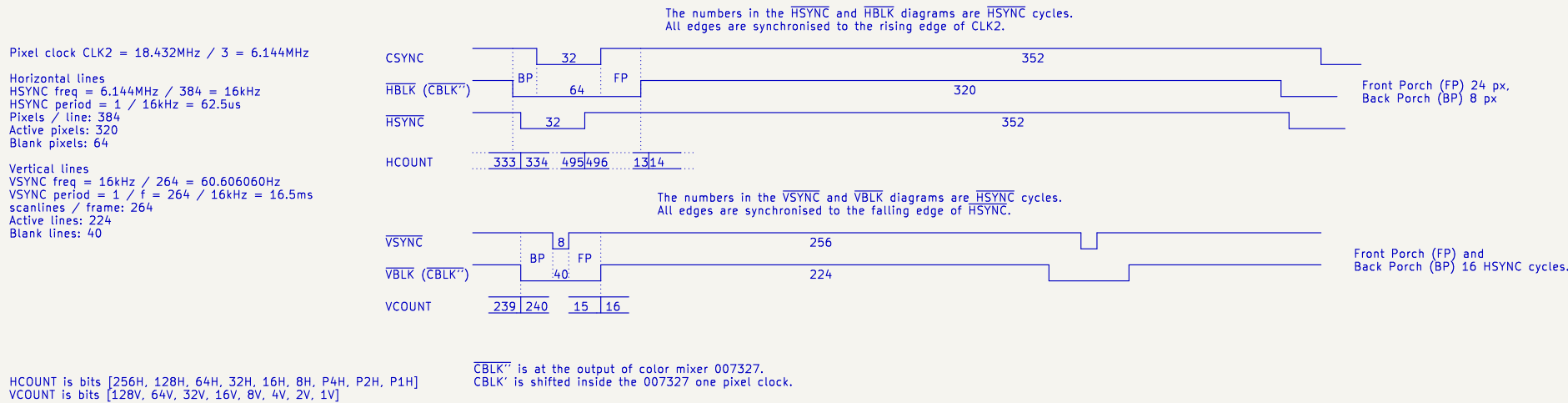
Horizontal signals



Vertical signals



Horizontal and vertical synch timing diagrams



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Konami GX687 / GX785

Sheet: /Timing diagrams/

File: timing_diagrams.kicad_sch

Title: Twin 16

Size: A3

Date: 2024-09-08

Rev:

KiCad E.D.A. 8.0.9

Id: 15/17

